

# RC Circuits Data Collection

Table 1: RC Circuit 1 (from First Semester)

|                      | $V_{DC}$    | $V_{R1}$    | $V_{R2}$    | $V_{R3}$      | $V_{R4}$    | $V_{OUT}$   |
|----------------------|-------------|-------------|-------------|---------------|-------------|-------------|
| <i>Calculated DC</i> |             |             |             |               |             |             |
| <i>Measured DC</i>   |             |             |             |               |             |             |
|                      | $V_{GENpp}$ | $V_{OUTpp}$ | $V_{R1pp}$  | $V_{R2pp}$    | $V_{R3pp}$  | $V_{R4pp}$  |
| <i>Calculated AC</i> |             |             |             |               |             |             |
| <i>Measured AC</i>   |             |             |             |               |             |             |
|                      | $FC_L Sine$ | $FC_H Sine$ | <i>Tilt</i> | $Time_{Rise}$ | $FC_L Sqr.$ | $FC_H Sqr.$ |
| <i>Calculated FC</i> |             |             |             |               |             |             |
| <i>Measured FC</i>   |             |             |             |               |             |             |

Table 2: Cycles to Stabilization

|                   | <i>Cycles</i> | $V_{MAX}$ | $V_{MIN}$ | <i>AWV</i> |
|-------------------|---------------|-----------|-----------|------------|
| <i>Calculated</i> |               |           |           |            |
| <i>Measured</i>   |               |           |           |            |

Table 3: RC Integration

|                   | $C$ | <i>Cycles</i> | $V_{MAX}$ | $V_{MIN}$ | <i>AWV</i> |
|-------------------|-----|---------------|-----------|-----------|------------|
| <i>Calculated</i> |     |               |           |           |            |
| <i>Measured</i>   |     |               |           |           |            |

Table 4: RC Differentiation

|                   | $PW$         | $Freq$       | <i>Cycles</i> | $V_{MAX}$    | $V_{MIN}$    |
|-------------------|--------------|--------------|---------------|--------------|--------------|
| <i>Calculated</i> |              |              |               |              |              |
| <i>Measured</i>   |              |              |               |              |              |
|                   | $V_{R1\tau}$ | $V_{R2\tau}$ | $V_{R3\tau}$  | $V_{R4\tau}$ | $V_{R5\tau}$ |
| <i>Calculated</i> |              |              |               |              |              |
| <i>Measured</i>   |              |              |               |              |              |
|                   | $V_{C1\tau}$ | $V_{C2\tau}$ | $V_{C3\tau}$  | $V_{C4\tau}$ | $V_{C5\tau}$ |
| <i>Calculated</i> |              |              |               |              |              |
| <i>Measured</i>   |              |              |               |              |              |

Table 5: RC Circuits Sine-wave Instantaneous Voltages Part 1

|                   | $V_{GEN}$ | $C$      | $R$        | $F_C$ | $V_C \angle^\circ$ | $V_R \angle^\circ$ |
|-------------------|-----------|----------|------------|-------|--------------------|--------------------|
| <i>Calculated</i> | $10V_P$   | $1\mu F$ | $1K\Omega$ |       |                    |                    |
| <i>Measured</i>   |           |          |            |       |                    |                    |

Table 6: RC Circuits Sine-wave Instantaneous Voltages Part 2

|                   | $Time$ | $V_{GEN}$<br><i>Calc.</i> | $V_{GEN}$<br><i>Meas.</i> | $V_R$<br><i>Calc.</i> | $V_R$<br><i>Meas.</i> | $V_C$<br><i>Calc.</i> | $V_C$<br><i>Meas.</i> |
|-------------------|--------|---------------------------|---------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <i>Reference</i>  | $0mS$  | $0V$                      | $0V$                      |                       |                       |                       |                       |
| <i>10% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>20% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>30% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>40% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>50% cycle</i>  |        | $0V$                      | $0V$                      |                       |                       |                       |                       |
| <i>60% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>70% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>80% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>90% cycle</i>  |        |                           |                           |                       |                       |                       |                       |
| <i>100% cycle</i> |        |                           |                           |                       |                       |                       |                       |